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Analyzing Leadership Competency in Driving Digital Transformation Within Manufacturing Enterprises Using a Spherical Z-Number PROMETHEE II Method

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Abstract Digital transformation of manufacturing enterprises demands a fundamental shift in leadership, requiring leaders to leverage digital technologies to drive organizational innovation. Leadership is a critical driver of digital transformation, and analyzing leadership competency is central to enhancing capabilities. Nonetheless, little research has explored this area through the lens of decision-making methodology. Hence, this study presents a hybrid decision-making framework designed to analyze the leadership competencies of manufacturing enterprises undergoing digital transformation. This framework is based on the Spherical Z-number (SZN) preference ranking organization method for enrichment evaluation (PROMETHEE) II methodology, which includes three phases. First, the group decision matrix is generated based on the SZN Relative Closeness Coefficient approach with the ordered weighted average operator. It describes more information than previous fuzzy sets through three evaluation branches and considers the reliability of experts. Second, the SZN LOPCOW (logarithmic percentage change-driven objective weighting) method for calculating criteria weights handles the structural inconsistencies of the criteria. Third, the SZN PROMETHEE II methodology is presented to obtain the complete priority of alternatives through pairwise comparisons and predefined preference functions. It also evaluates the deviation of any pair of alternatives using a simplified computational procedure. Moreover, an experimental case study of an automobile manufacturing enterprise in China is presented to verify the practicality of the proposed methodology. According to the results, "Emotional balance (ζ_{16})" in the dimension of "Digital ethical awareness" is the main leadership competency with the largest value, followed by the leadership competency "Digital empathy ability (ζ_{13})." Within the analysis of leadership competency, the candidate χ_5 has the highest priority compared to others. Finally, the sensitivity and comparative analyses demonstrate the robustness and validity of the presented SZN PROMETHEE II methodology.

Keywords Leadership competency · Multi-Criteria Decision-Making · Spherical Z-number · PROMETHEE II

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1 Introduction

The rapid advancement of digital technologies, including big data, artificial intelligence, mobile Internet, cloud computing, 5G, and the Internet of Things, has transformed industries and reshaped the organizational landscape. Manufacturing enterprises are undergoing digital transformation in the digital era [1]. Digital transformation in manufacturing enterprises provides numerous opportunities. For example, digitalization enables manufacturing enterprises to gain market competitiveness by constantly innovating, enhancing efficiency, and

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optimizing production processes [2]. Additionally, it accelerates the design and commercialization of products, enabling manufacturing enterprises to meet consumer demands more quickly. Manufacturing enterprises face many challenges despite these opportunities during their digital transformation [1]. Digital transformation has shifted the reform of systems, management, and operations of manufacturing enterprises. Traditional leadership styles are no longer suitable for current organizational development. Manufacturing leaders must shift their mindset, adapt management practices, and leverage digital technologies to ensure organizational efficiency [3]. Successful digital transformation requires leadership competency that can foster organizational innovation and structural reform in today's digital landscape [4]. Leadership competency in driving digital transformation refers to leaders' ability to leverage technological tools to enhance decision-making, efficiency, and organizational performance in response to a rapidly changing environment [5]. It is a digitally enabled, change-oriented, cross-hierarchical leadership strategy. Leadership competency drives manufacturing enterprises to employ technologies, foster employee innovation, and optimize operational procedures [6]. Leadership plays a pivotal role in the digital transformation of a manufacturing enterprise. Analyzing leadership competency is the core issue in enhancing capabilities and influencing the efficiency of digital transformation.

The external environment faced by manufacturing enterprises has become volatile, complex, and uncertain. For example, increasing the complexity of production factors, diversifying internal correlations, and growing demand uncertainty [3,7]. Facing these challenges, analyzing leadership competency in driving digital transformation within manufacturing enterprises is hindered by the limited availability of evaluation data. Fuzzy set theory is used to solve this problem in existing research on competency assessment and manufacturing enterprise management [8,9]. Kononenko et al. [10] employed an Intuitionistic fuzzy set to cover uncertain information while assessing the competency of applicants for a project team. Sumrit [11] applied the Pythagorean fuzzy set (PFS) to evaluate the industrial IoT in manufacturing enterprises. Görçün et al. [12] used the Fermatean fuzzy set to analyze 3D Printer technology for the sustainable manufacturing industry. The above fuzzy sets describe complex information via membership and non-membership functions. However, these fuzzy sets are restricted when capturing a wider range of uncertain data. As a generalization of the above fuzzy sets, the Spherical fuzzy set (SFS) proposed by .Kahraman and GÜNDOĞDU [13] provides a larger preference domain for experts [14]. Experts can define the hesitancy information of an alternative based on a criterion. Although SFS is a powerful tool for modeling evaluation information in the decision-making process, it cannot adequately express the reliability of experts. The Spherical Z-number (SZN), introduced by Ashraf et al. [15], combines the advantages of SFS and Z-number. It expresses more information with a three-branch evaluation than existing fuzzy sets and reflects the expert's reliability during decision-making [16]. SZN has been employed to solve many decision-making problems. Examples include green product design [17], assessment of economic development policies [15], and evaluation of climate change [18]. Despite these promising findings, little research has focused on leadership competency analysis using SZN, especially for manufacturing enterprises undergoing digital transformation. Therefore, this paper presents an SZN-based decision-making method for analyzing leadership competency in driving digital transformation within manufacturing enterprises.

Analyzing the leadership competency of manufacturing in driving digitalization requires consideration of criteria from various aspects. Examples include digital strategic vision [19], digital technology learning [20], digital implementation [3], and digital ethical empathy [21]. Due to the concern that multiple criteria feed into the decision-making process, the existing literature on leadership competency analysis regards it as a Multi-Criteria Decision-Making (MCDM) issue [22]. The MCDM method can solve problems with conflicting goals and individual perspectives under uncertainty [23,24]. The PROMETHEE (Preference Ranking Organization Method for Enrichment Evaluation) method, as a practical MCDM approach, is an outranking approach that can rank a finite set of alternatives under usually conflicting criteria. The PROMETHEE method identifies the optimal option through pairwise comparisons using six predefined preference functions to evaluate the deviations between alternatives [25]. However, the conventional PROMETHEE I method determines alternative priorities using outgoing and incoming flows, which is a partial ranking. The PROMETHEE II method can obtain a complete ranking of alternatives rather than a partial ranking based on net flows [25]. Studies utilizing the PROMETHEE II method are rich in other decision-making problems, but remain limited in their application to competency analysis under uncertain contexts. Hence, this study introduces a novel SZN PROMETHEE II methodology to assess leadership competency in driving digital transformation within manufacturing enterprises.

1.1 Motivations

- The rise of the digital age has created a new business environment for manufacturing companies, bringing opportunities and challenges. Analyzing leadership competency in driving digital transformation provides critical insight for manufacturing enterprises. Such analysis helps improve organizational flexibility and resilience, enabling companies to better handle the challenges of a rapidly changing digital world.

- Digital transformation introduces complexity and uncertainty into manufacturing enterprises' external environment. To respond effectively, leaders must adapt their strategies to meet these evolving challenges. However, assessing leadership competencies within this context remains difficult due to limited data availability.
- Analyzing leadership competencies in digitally transforming manufacturing enterprises must account for uncertain environments and multidimensional criteria. It requires a comprehensive decision-making framework. The framework should ensure the reliability of evaluation data under uncertainty and incorporate expert preferences for practical relevance.
- A structured indicator system is essential for analyzing leadership competencies in digital transformation. Such a system should integrate multiple critical leadership dimensions pertinent to manufacturing digitalization. Moreover, it must be designed to assess competencies aligned with current demands, thereby enhancing its practical applicability.

1.2 Novelty and Contributions

- A novel SZN PROMETHEE II framework for leadership competency analysis under digital transformation: This study presents the first integration of SZNs with the PROMETHEE II approach. This novel framework contributes by introducing a pairwise comparison mechanism for preference modelization within SZN environment, a feature that distinguishes it from conventional MCDM methods like TOPSIS [26], COPRAS [27], WASPAS [28], and EDAS [29] commonly used in competence evaluation. It offers a refined and reliability-sensitive decision-making tool designed to handle the complex uncertainties inherent in leadership assessments during digital transformation.
- A new information aggregation methodology incorporating reliability and expert weighting: Based on the traditional SZN aggregation techniques, this study introduces an integrated approach combining the Relative Closeness Coefficient (RCC) method and the Ordered Weighted Average (OWA) operator. It captures the three-dimensional uncertainty of SZNs and determines expert weights adaptively based on the ordered positions of assessments. It also reduces subjectivity in expert weighting and enhances the robustness of group decision matrices.
- A novel MCDM approach for leadership competency analysis in digital transformation. This study proposes an integrated method combining SZN LOPCOW (logarithmic percentage change-driven objective weighting) for objective criterion weighting, addressing issues such as dimensional disparities. The SZN PROMETHEE II for systematic ranking via pairwise comparisons with simplified preference functions. It represents an early application of SZN-based PROMETHEE II to leadership competence analysis.
- A systematic index system for leadership competency analysis of manufacturing enterprises undergoing digital transformation: The system consists of four dimensions that together include sixteen criteria. This framework provides a structured basis for analyzing leadership capabilities in manufacturing enterprises undergoing digital transformation.

1.3 Organization of the Study

This paper has six sections. Section 2 outlines an overview of the related research. Section 3 recalls fundamental concepts and preliminaries. Section 4 describes the proposed methodology, which consists of three stages. Section 5 presents an experimental case study, including a sensitivity analysis, comparative analysis, and managerial implications. Section 6 concludes the paper, explaining the outcomes and outlining future explorations.

2 Literature Review

This section reviews the related research of this study, including leadership analysis within digital transformation, the development of competency evaluation methods, the application of the SZN method, and the application of the fuzzy PROMETHEE II method.

2.1 Leadership Analysis Within Digital Transformation

While analyzing leadership within digital transformation, some studies have used a literature review focusing on its concepts and characteristics. For example, Malik et al. [30] examined the characteristics of digital leadership and the role of leaders in digitalization through an integrated review. Lin [4] clarified the concepts and the structure of digital leadership through a systematic literature review. Furthermore, most of these studies utilized the empirical method. Pham et al. [31] empirically discussed the moderating role of empowering leadership and the connection between employee outcomes and digital-oriented training programs in remote work contexts. Xia et al. [32] empirically examined the impact of digital leadership by lead entrepreneurs on digital entrepreneurial success. Esangbedo et al. [33] analyzed the relationship between supply chain leadership and the sustainable performance of logistics enterprises through an empirical method. Bilal et al. [34] empirically identified the

key factors that impact the digital transformation of manufacturing enterprises. Chong and Zainal [35] empirically examined the impacts of employee transformational leadership and digital literacy on job performance. Zada et al. [36] empirically investigated the effects of leadership on digital transformation, sustainable performance, and green innovation. Kludacz-Alessandri et al. [37] proposed a framework to assess how digital transformation leadership influences digital intensity for healthcare entities. Probojakti et al. [38] analyzed the role of transformational leadership and digital transformation in improving organizational agility and corporate resiliency in banking. Only a few studies have used MCDM methods. Yoo et al. [39] employed the fuzzy DEMATEL approach to identify the crucial factors for actualizing digital leadership.

2.2 Development of the Competencies Analysis Methods

Prior literature has extensively applied the MCDM methods to competency evaluation issues across diverse domains. For example, Bekesiene et al. [26] employed the fuzzy AHP (analytical hierarchy process)-TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution) method to identify the primary competencies required for soldier resilience training. Nalbant [40] applied the Interval Type 2 fuzzy TOPSIS method to select the project managers, considering their technical capabilities, managerial skills, and communication competence. Rahadian et al. [28] analyzed the fintech competencies in emerging markets using the WASPAS (Weighted Aggregated Sum Product Assessment) method with SFS. Masoomi et al. [27] introduced the fuzzy COPRAS method based on green capabilities for the renewable energy supply chain. Feng et al. [29] employed the EDAS (Evaluation based on Distance from Average Solution) method with PFS to assess the big data competence for manufacturing enterprises. Some researchers considered the interrelationship and the hierarchical structure of competencies. Bekesiene et al. [41] developed the DEMATEL (Decision-Making Trial and Evaluation Laboratory) method with trapezoidal fuzzy numbers to evaluate soldiers' mental resilience competencies. Wang et al. [22] employed the DEMATEL method with Z-numbers to examine the leadership competencies of the financial industry during digital transformation. Yao et al. [42] analyzed the necessary competence and its hierarchy structure in sustainable project management by the fuzzy ISM–MICMAC (interpretive structural modeling–matrix impact cross-reference multiplication applied to classification) method. Some studies considered the preference of experts during competence evaluation. Further, Wang [43] established a new framework based on the fuzzy AHP-PROMETHEE and employed it in captain selection problems. The summary of the decision-making methods for competence evaluation is shown in Table 1.

2.3 Spherical Z-Number Method

SZN considers a wider range of uncertain information and its reliability for evaluation, which is a helpful method for evaluating problems with complex contexts. This subsection concludes the recent research on SZN-based decision-making methodologies. Ashraf et al. [15] initially introduced a decision-making model based on the SZN method, discussing its operational laws and fusion operators. Ashraf et al. [44] evaluated the appropriate implant design for a biomedical knee via the SZN-based operator. This was applied to assess economic development policies. Huang et al. [17] proposed a comprehensive ARAS (additive ratio assessment) method with SZNs for solving green product design issues. Wu et al. [16] defined the SZN-based group decision-making model using the prospect theory, considering the irrational behavior of experts. Ashraf et al. [18] investigated the SZN-based extended TODIM (an acronym in Portuguese for Interactive and Multi-criteria Decision-Making) method to assess climate change, developing operational rules and aggregation operators. Nguyen [45] developed a comprehensive group of expert consensus tools based on the SZN PHI model. Niu [46] applied an SZN-based MARCOS (measurement of alternatives and ranking according to the compromise solution) method to assess the performance of English teachers. The progress of the research on the SZN method is shown in Table 2.

2.4 The PROMETHEE II Method

The PROMETHEE II method can determine the final ranking of experts, considering the deviation of each pair of alternatives, and is widely applied in research. Seikh and Mandal [47] employed the PROMETHEE II method within Interval-Valued fuzzy settings to evaluate the most suitable organization for managing biomedical waste. Akram et al. [25] developed a PROMETHEE II method with SFS and applied it to select the most appropriate site for constructing the Fangcang shelter hospital. Oubahman and Duleba [48] developed a novel model utilizing the fuzzy PROMETHEE II method to analyze the choice of public transportation modes. Akram et al. [49] proposed a multi-polar fuzzy N-soft PROMETHEE approach to enhance digital economic growth. Wang et al. [50] analyzed the barriers to adopting a resilient food supply chain based on the PROMETHEE II method within Interval-Valued fuzzy settings. Bakas et al. [51] established a hybrid Intuitionistic Fuzzy PROMETHEE II method for water source management. Mohammed et al. [52] proposed a hybrid decision model based on the PROMETHEE with Spherical cubic fuzzy to evaluate the

Table 1 Summary of the decision-making methods for competence analysis

Authors	Fuzzy set	Method	Applied area	Reliability	Preference
Bekesiene et al. [26]	Triangular fuzzy numbers	AHP-TOPSIS	Military organization	×	×
Nalbant [40]	Interval Type 2 fuzzy	TOPSIS	Human resource management	×	×
Rahadian et al. [28]	SFS	WASPAS	Fintech competency for emerging markets	×	×
Masoomi et al. [27]	Triangular fuzzy numbers	WASPAS-COPRAS	Energy supply chain	×	×
Feng et al. [29]	PFS	EDAS	manufacturing supply chains	×	×
Bekesiene et al. [41]	Trapezoidal fuzzy numbers	DEMATEL	Military resilience	×	×
Wang et al. [22]	Z-numbers	DEMATEL	Financial industry	√	×
Yao et al. [42]	Triangular fuzzy numbers	ISM–MICMAC	Sustainable project management	×	×
Wang [43]	Intuitionistic rough fuzzy	AHP-PROMETHEE	Team captain selection	×	√
The proposed method	SZN	LOPCOW-PROMETHEE II	Leadership competence in driving digital transformation	√	√

Table 2 Research progress on the SZN method

Authors	Method	Applied area
Ashraf et al. [18]	Sugeno–Weber model	Climate change
Ashraf et al. [44]	Neutrality aggregation operators	Knee osteoarthritis transplant
Niu [46]	MARCOS	Education performance
Huang et al. [17]	ARAS	Green product design
Wu et al. [16]	Prospect theory	Shared e-bike recycling supplier selection
Nguyen [45]	PHI model	Logistics supply chain management
Ashraf et al. [15]	Extended TODIM	Economic development policies
The proposed method	PROMETHEE II	Leadership competence in driving digital transformation

cloud fault tolerance mechanism. Li et al. [53] extended the PROMETHEE method into the Z-number settings and discussed its ordering rules.

2.5 Research Gaps

Based on the discussion above, we conclude with several limitations of the existing research.

- Existing studies have extensively investigated leadership in digital transformation across various application domains, such as government governance [54], education [55], public healthcare, social media [56], and enterprises [6]. Current research on leadership within digital transformation has primarily followed two methodological approaches: conceptual analyses through literature reviews and empirical investigations of organizational impacts. However, few studies have discussed this topic using MCDM methods, particularly in the context of leadership competency analysis and its role in driving digital transformation. Also, it lacks a systematic index for analyzing the leadership competency in driving digital transformation for organizations.
- Existing research has demonstrated the efficacy of MCDM methods for competency evaluation across various domains, employing diverse fuzzy sets (Triangular fuzzy numbers [26], Intuitionistic rough fuzzy [43], PFS [29], SFS [28], and so on) and hybrid techniques (TOPSIS [26], WASPAS [28], COPRAS [27], EDAS [29], and so on). Nevertheless, current methods are constrained by three limitations: inadequate consideration of reliability factors in most methods (only Wang et al. [22] incorporated this aspect); limited integration of expert preferences (only Wang [43] addressed this dimension); and absence of applications to leadership competency within digital transformation.
- While SZN-based MCDM methods have demonstrated effectiveness in handling complex uncertainties across diverse domains, existing applications remain constrained to technical and operational decision con-

texts (e.g., medical implants [44] and supply chain management [45]). Current implementations exhibit three limitations: (1) predominant focus on product service evaluation rather than human competency assessment, (2) underutilization in organizational leadership contexts, and (3) absence of integration with the PROMETHEE II method.

- The study body employing the fuzzy PROMETHEE II method is primarily focused on waste management [47], supply chain [50], and public transportation [48]. However, competency assessment, especially leadership competency within digital transformation, has received less attention in prior research.

3 Preliminaries

In this section, we recall some basic concepts of SFS and SZN.

Definition 1 [16]. For any universal set R, the SFS is defined as follows:

$$\mathrm{H} = \{(f, \mu(f), \nu(f), \gamma(f)) \,|\, f \in R\}, \tag{1}$$

where $\mu(f) \in [0, 1]$,$\nu(f) \in [0, 1]$,$\gamma(f) \in [0, 1]$ represent membership degree, neutrality degree, and non-membership degree, respectively, satisfying $0 \leq \mu^2(f) + \nu^2(f) + \gamma^2(f) \leq 1$. The refusal degree of f in R is defined by $\pi(f) = \sqrt{1 - \left(\mu^2(f) + \nu^2(f) + \gamma^2(f)\right)}$.

Definition 2 [16]. Let $\mathrm{H}_1 = \left(\mu_1, \nu_1, \gamma_1\right)$ and $\mathrm{H}_2 = (\mu_2, \nu_2, \gamma_2)$ be two spherical fuzzy numbers (SFNs), their corresponding operations are defined below.

$$\mathrm{H}_1 \oplus \mathrm{H}_2 = \left\{\left(\mu_1^2 + \mu_2^2 - \mu_1^2\mu_2^2\right)^{\frac{1}{2}}, \nu_1\nu_2, \gamma_1\gamma_2\right\}, \tag{2}$$

$$\mathrm{H}_1 \otimes \mathrm{H}_2 = \left\{\mu_1\mu_2, \left(\nu_1^2 + \nu_2^2 - \nu_1^2\nu_2^2\right)^{\frac{1}{2}}, \left(\gamma_1^2 + \gamma_2^2 - \gamma_1^2\gamma_2^2\right)^{\frac{1}{2}}\right\}, \tag{3}$$

$$\lambda \cdot \mathrm{H}_1 = \left\{\left(1 - \left(1 - \mu_1^2\right)^{\lambda}\right)^{\frac{1}{2}}, \nu_1^{\lambda}, \gamma_1^{\lambda}\right\}, \lambda > 0, \tag{4}$$

$$\mathrm{H}_1^{\lambda} = \left\{\mu_1^{\lambda}, \left(1 - \left(1 - \nu_1^2\right)^{\lambda}\right)^{\frac{1}{2}}, \left(1 - \left(1 - \gamma_1^2\right)^{\lambda}\right)^{\frac{1}{2}}\right\}, \lambda > 0. \tag{5}$$

Definition 3 [57]. Let $\mathrm{H} = (\mu, \nu, \gamma)$ be an SFN, the score and accuracy functions are defined as follows:

$$\mathrm{s}(\mathrm{H}) = \frac{1}{2}\left(1 + \mu^2 - \nu^2 - \gamma^2\right), \tag{6}$$

$$W(\mathrm{H}) = \mu^2 + \nu^2 + \gamma^2. \tag{7}$$

Definition 4 [16]. For two SFNs $\mathrm{H}_1 = \left(\mu_1, \nu_1, \gamma_1\right)$ and $\mathrm{H}_2 = (\mu_2, \nu_2, \gamma_2)$, the distance between them is defined as below.

$$D(\mathrm{M}_1, \mathrm{M}_2) = \frac{1}{2}\left(\left|\mu_1^2 - \mu_2^2\right| + \left|\nu_1^2 - \nu_2^2\right| + \left|\gamma_1^2 - \gamma_2^2\right| + \left|\pi_1^2 - \pi_2^2\right|\right), \tag{8}$$

where π_1 ,π_2 is the refusal degree of H_1 and H_2, respectively, and satisfies $D(\mathrm{H}) \in [0, 1]$.

Definition 5 [57]. For two SFNs $\mathrm{H}_1 = \left(\mu_1, \nu_1, \gamma_1\right)$ and $\mathrm{H}_2 = (\mu_2, \nu_2, \gamma_2)$, their corresponding mathematical formulas between them are defined by

(1) If $\mathrm{s}(\mathrm{H}_1) < \mathrm{s}(\mathrm{H}_2)$, then $\mathrm{H}_1 < \mathrm{M}_2$.
(2) If $\mathrm{s}(\mathrm{M}_1) = \mathrm{s}(\mathrm{M}_2)$, then

① If $W(\mathrm{H}_1)$, then $\mathrm{H}_1 < \mathrm{H}_2$.
② If $W(\mathrm{H}_1) = W(\mathrm{H}_2)$, then $\mathrm{H}_1 \sim \mathrm{H}_2$.

Definition 6 [16]. Let $Z_{SZN} = (A, B)$ be a SZN, which is a fuzzy number calculated by integrating the constraint function A and the reliability function B, where both A and B are SFNs. Then, the SZN is denoted as.

$$Z_{SZN} = (A, B) = \{(\mu_A(f), \nu_A(f), \gamma_A(f)), (\mu_B(f), \nu_B(f), \gamma_B(f))\}. \tag{9}$$

Definition 7 [16]. To reduce computational complexity, the A and B in SZN are combined. The simplified SZNs are still SFNs that maintain their corresponding properties. The simplified SZN can be expressed as follows:

$$Z'_{SZN} = (\mu_{SZN}(f), \nu_{SZN}(f), \gamma_{SZN}(f)) = \begin{pmatrix} \mu_A(f)\mu_B^{\frac{1}{2}}(f), \left(\nu_A^2(f) + \left(1 - \left(1 - \nu_B^2(f)\right)^{\frac{1}{2}}\right) - \nu_A^2(f)\left(1 - \left(1 - \nu_B^2(f)\right)^{\frac{1}{2}}\right)\right)^{\frac{1}{2}}, \\ \left(\gamma_A^2(f) + \left(1 - \left(1 - \gamma_B^2(f)\right)^{\frac{1}{2}}\right) - \gamma_A^2(f)\left(1 - \left(1 - \gamma_B^2(f)\right)^{\frac{1}{2}}\right)\right)^{\frac{1}{2}} \end{pmatrix}. \tag{10}$$

Definition 8 [58]. The simplified SZN OWA operator is denoted by

$$SZNOWA(Z_1, Z_2, ..., Z_q) = \left\{\left(1 - \prod_{t=1}^{q}\left(1 - \mu_{l(t)}^2\right)^{\omega_t}\right)^{\frac{1}{2}}, \prod_{t=1}^{q}\nu_{l(t)}^{\omega_t}, \prod_{t=1}^{q}\gamma_{l(t)}^{\omega_t}\right\}; t = 1, \cdots, q, \tag{11}$$

where $Z_{l(t)}$ is the tth largest value consequently by total order $Z_{l(1)} \geq Z_{l(2)} \geq \cdots \geq Z_{l(q)}$, and $\omega_t = \left(\omega_1, \omega_2, \cdots \omega_q\right)^T$

is an associated vector of the operator, $\omega_t \in [0, 1]$ and $\sum_{t=1}^{q} \omega_t = 1$.

$$M_{ij}^t = \begin{array}{c} \\ \chi_1 \\ \chi_2 \\ \vdots \\ \chi_m \end{array} \begin{array}{c} \begin{array}{ccc} \zeta_1 & \cdots & \zeta_n \end{array} \\ \begin{bmatrix} \left(\left(\mu_{A_{11}^t}(f), \nu_{A_{11}^t}(f), \gamma_{A_{11}^t}(f)\right), \left(\mu_{B_{11}^t}(f), \nu_{B_{11}^t}(f), \gamma_{B_{11}^t}(f)\right)\right) & \cdots & \left(\left(\mu_{A_{1n}^t}(f), \nu_{A_{1n}^t}(f), \gamma_{A_{1n}^t}(f)\right), \left(\mu_{B_{1n}^t}(f), \nu_{B_{1n}^t}(f), \gamma_{B_{1n}^t}(f)\right)\right) \\ \left(\left(\mu_{A_{21}^t}(f), \nu_{A_{21}^t}(f), \gamma_{A_{21}^t}(f)\right), \left(\mu_{B_{21}^t}(f), \nu_{B_{21}^t}(f), \gamma_{B_{21}^t}(f)\right)\right) & \cdots & \left(\left(\mu_{A_{2n}^t}(f), \nu_{A_{2n}^t}(f), \gamma_{A_{2n}^t}(f)\right), \left(\mu_{B_{2n}^t}(f), \nu_{B_{2n}^t}(f), \gamma_{B_{2n}^t}(f)\right)\right) \\ \vdots & \ddots & \vdots \\ \left(\left(\mu_{A_{m1}^t}(f), \nu_{A_{m1}^t}(f), \gamma_{A_{m1}^t}(f)\right), \left(\mu_{B_{m1}^t}(f), \nu_{B_{m1}^t}(f), \gamma_{B_{m1}^t}(f)\right)\right) & \cdots & \left(\left(\mu_{A_{mn}^t}(f), \nu_{A_{mn}^t}(f), \gamma_{A_{mn}^t}(f)\right), \left(\mu_{B_{mn}^t}(f), \nu_{B_{mn}^t}(f), \gamma_{B_{mn}^t}(f)\right)\right) \end{bmatrix}_{m \times n} \end{array} \tag{12}$$

4 Research Methodology

Here, we introduce an MCDM method that utilizes the PROMETHEE II and LOPCOW methods within the SZN framework. There are three primary phases in the introduced methodology. First, we construct a group decision matrix. We combine the SZN RCC method with the OWA operator to compute the expert weights and aggregate the experts' information. It determines expert weights by addressing the limitations of existing distance measures, particularly their inability to distinguish between experts in certain situations [59]. It also reflects the significance degrees of both the SZN expert and the ordered position of the expert. Second, the criterion weights are computed. Using the SZN LOPCOW method, we determine the weights of these criteria. It defines the effective criteria weights by removing various problems from different criteria structures, such as dimensional disparities and negative values [60]. Third, the alternatives are ranked. The SZN PROMETHEE II method is utilized to rank the alternatives, considering the deviation between any pair of alternatives and the modeling preference. The methodology steps are illustrated in Fig.1, and the stepwise procedure is summarized in the algorithm presented in Fig. 2.

Stage 1: Development of the group evaluation matrix utilizing the SZN RCC method in conjunction with the OWA operator.

We employ the SZN RCC method to determine the experts' weights. Furthermore, the OWA operator is used to generate the group evaluation matrix using the following steps:

Step 1.1: Construct an individual evaluation matrix for experts. Let $X = \{\chi_i \,|\chi_1, \chi_2, \cdots, \chi_m\}$ be the set of alternatives, $\Upsilon = \left\{\zeta_j \,|\zeta_1, \zeta_2, \cdots, \zeta_n\right\}$ be the set of criteria, and $E = \left\{\varepsilon_t \,|\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_q\right\}$ be the set of experts. Then, the expert ε_t gives the evaluation information of the alternative χ_i under the criteria ζ_j according to the linguistic scale shown in Table 3. Consequently, the individual evaluation matrix $M^t = \left[M_{ij}^t\right]_{m \times n}$ for the tth expert is represented as follows:

where $M_{ij}^t = \left(\left(\mu_{A_{ij}^t}(f), \nu_{A_{ij}^t}(f), \gamma_{A_{ij}^t}(f)\right), \left(\mu_{B_{ij}^t}(f), \nu_{B_{ij}^t}(f), \gamma_{B_{ij}^t}(f)\right)\right)$ is a SZN, denoting the evaluation information of the ith alternative under the jth criteria given by the expert ε_t. Then, transform the SZNs into the simplified SZNs according to Eq. (10).

Step 1.2: Determine the reference preference data. Based on the individual evaluation matrix $M^t = \left[M_{ij}^t\right]_{m \times n}$, the positive and negative reference preference data are shown as follows:

$$M_t^+ = \max_t \left\{\max M_{ij}^t\right\}, \tag{13}$$

$$M_t^- = \min_t \left\{\min M_{ij}^t\right\}, \tag{14}$$

where M_t^+ and M_t^- are two simplified SZNs, expressed as $M_t^+ = \left(\mu_t^+, \nu_t^+, \gamma_t^+\right)$ and $M_t^- = \left(\mu_t^-, \nu_t^-, \gamma_t^-\right)$, respectively.

Step 1.3: Calculate the distances. According to the preference data and Eq. (8), the positive and negative distance measures are calculated as follows:

$$\begin{aligned} dp_t = \sum_{i=1}^{m}\sum_{j=1}^{n}\Bigg(\frac{1}{2}\Bigg(&\left|\left(\mu_{ij}^t\right)^2 - \left(\mu_t^+\right)^2\right| \\ &+ \left|\left(\nu_{ij}^t\right)^2 - \left(\nu_t^+\right)^2\right| \\ &+ \left|\left(\gamma_{ij}^t\right)^2 - \left(\gamma_t^+\right)^2\right| + \left|\left(\pi_{ij}^t\right)^2 - \left(\pi_t^+\right)^2\right|\Bigg)\Bigg), \end{aligned} \tag{15}$$

$$\begin{aligned} dn_t = \sum_{i=1}^{m}\sum_{j=1}^{n}\Bigg(\frac{1}{2}\Bigg(&\left|\left(\mu_{ij}^t\right)^2 - \left(\mu_t^-\right)^2\right| \\ &+ \left|\left(\nu_{ij}^t\right)^2 - \left(\nu_t^-\right)^2\right| + \left|\left(\gamma_{ij}^t\right)^2 - \left(\gamma_t^-\right)^2\right| \\ &+ \left|\left(\pi_{ij}^t\right)^2 - \left(\pi_t^-\right)^2\right|\Bigg)\Bigg). \end{aligned} \tag{16}$$

Step 1.4: Compute the decision rating.

$$dr_t = \frac{dn_t}{dp_t + dn_t}. \tag{17}$$

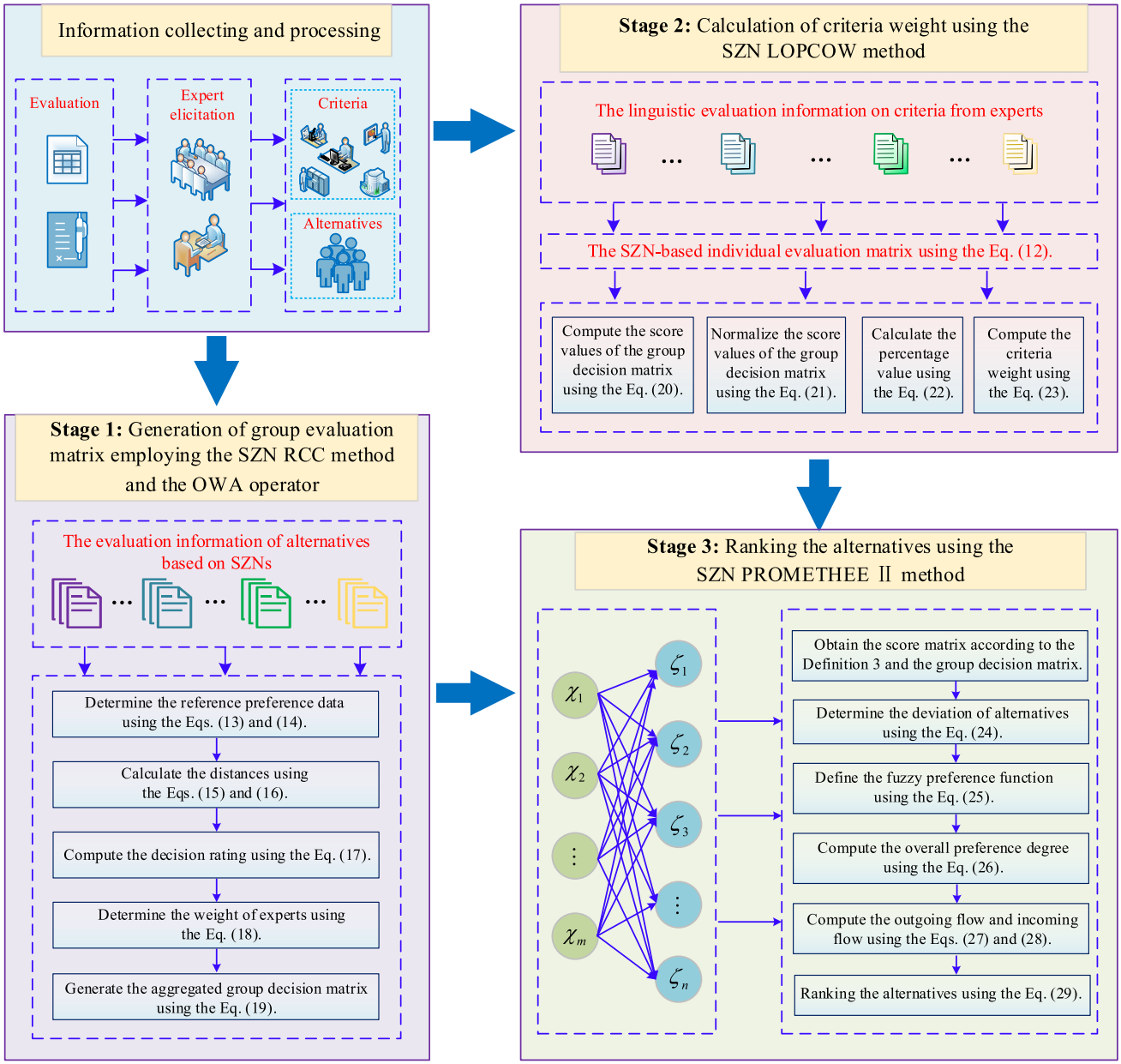

Fig. 1 The flowchart of the SZN PROMETHEE II method

Step 1.5: Determine the weight of experts.

$$\omega_t = \frac{dr_t}{\sum_{t=1}^{q} dr_t}. \tag{18}$$

Step 1.6: Generate the aggregated group decision matrix $\mathrm{Y} = [\mathrm{y}_{ij}]_{m\times n}$. Based on the OWA operator and expert's weights, the SZN-based aggregation approach is defined as follows:

$$\begin{aligned} y_{ij} &= SZNOWA\left(M_{ij}^1, M_{ij}^2, \ldots, M_{ij}^q\right) \\ &= \left\{\left(1-\prod_{t=1}^{q}\left(1-\mu^2_{l(Z_{ij}^t)}\right)^{\omega_t}\right)^{\frac{1}{2}}, \prod_{t=1}^{q} \nu^{\omega_t}_{l(M_{ij}^t)}, \prod_{t=1}^{q} \gamma^{\omega_t}_{l(M_{ij}^t)}\right\}; t = 1, \cdots, q \end{aligned}, \tag{19}$$

where $\left(l(M_{ij}^1), l(M_{ij}^2), \cdots, l(M_{ij}^q)\right)$ is the permutation of $\left(M_{ij}^1, M_{ij}^2, \cdots, M_{ij}^q\right)$ such that $l(M_{ij}^{t-1}) \geq l(M_{ij}^t)$ for all t; $l(M_{ij}^t)$ is the tth largest of M_{ij}^t; $\omega_t = \left(\omega_1, \omega_2, \cdots, \omega_q\right)^T$ is the weight of the expert ε_t, $\omega_t \in [0, 1]$ and $\sum_{t=1}^{q} \omega_t = 1$.

Stage 2: Calculation of criteria weight using the SZN LOPCOW method.

In this stage, we employ the SZN LOPCOW method to compute the weight of the criteria. The detailed computation steps are outlined as follows:

Fig. 2 The stepwise procedure of the proposed SZN PROMETHEE II method

Algorithm: The analyzing framework for leadership competency

Input: The set of alternatives $\chi_i\left(i=1,2,\cdots,m\right)$, a set of criteria $\zeta_j\left(j=1,2,\cdots,n\right)$ and a group of experts $\varepsilon_t\left(t=1,2,\cdots,q\right)$; the individual evaluation matrix $\mathrm{M}^t=\left[\mathrm{M}_{ij}^t\right]_{m\times n}$.

Output: The ranking of alternatives

Begin

Step 1. Form the individual evaluation matrix for experts $\mathrm{M}^t=\left[\mathrm{M}_{ij}^t\right]_{m\times n}$.

Step 2. for t to q

Compute the weight of experts ω_t using the Eqs. (13-18).

end for

Step 3. Form the aggregated group decision matrix $\mathsf{Y}=\left[\mathsf{y}_{ij}\right]_{m\times n}$ using the Eq. (19).

Step 4. for j to n

Compute the criteria weights w_j using the Eqs. (20-23).

Step 5. for j to n

Compute the score matrix $\mathrm{S}=\left[\mathsf{s}_{ij}\right]_{m\times n}$ according to the Definition 3.

Step 6. Compute the deviations of alternatives $d_j\left(\zeta_\alpha,\zeta_\beta\right)$ using the Eq. (24).

Step 7. Compute the fuzzy preference function $\mathsf{p}_j\left(\alpha,\beta\right)$ using the Eq. (25).

Step 8. Compute overall preference degree $\Pi\left(\alpha,\beta\right)$ using the Eq. (26).

Step 9. Compute the outgoing flow $\Phi^+\left(\alpha\right)$ and incoming flow $\Phi^-\left(\alpha\right)$ using the Eqs. (27) and (28).

Step 10. Compute the net flow $\Phi\left(\alpha\right)$ and ranking the alternatives using the Eq. (29).

End

Step 2.1: Compute the score values of the group decision matrix according to Eq. (6).

$$\mathsf{s}_{ij}=\frac{1}{2}\left(1+\mu_{ij}^2-\nu_{ij}^2-\gamma_{ij}^2\right). \tag{20}$$

Step 2.2: Normalize the score values of the group decision matrix.

$$\mathsf{N}_{ij}=\begin{cases}\dfrac{\max\limits_j \mathsf{s}_{ij}-\mathsf{s}_{ij}}{\max\limits_j \mathsf{s}_{ij}-\min\limits_j \mathsf{s}_{ij}}, & j\in j^+\\ \dfrac{\mathsf{s}_{ij}-\min\limits_j \mathsf{s}_{ij}}{\max\limits_j \mathsf{s}_{ij}-\min\limits_j \mathsf{s}_{ij}}, & j\in j^-\end{cases}, \tag{21}$$

where j^+ denotes the benefit criterion and j^- denotes the cost criterion.

Step 2.3: Calculate the percentage value.

$$\mathsf{P}_j=\left|\ln\left(\frac{\sqrt{\dfrac{\sum\limits_{i=1}^{m}\mathsf{N}_{ij}^2}{m}}}{\sigma_j}\right)\right|, \tag{22}$$

where σ_j is the deviation value of the criteria, and m is the number of alternatives.

Step 2.4: Compute the criteria weights.

$$_j=\frac{\mathsf{P}_j}{\sum\limits_{j=1}^{n}\mathsf{P}_j}. \tag{23}$$

Stage 3: Ranking alternatives using the SZN PROM ETHEE II method.

In this phase, we employ the SZN PROMETHEE II method to prioritize alternatives. The detailed steps are as follows.

Step 3.1: Obtain score matrix. Based on the score function in Definition 3 and the group decision matrix, we compute the score matrix $S=\left[\mathsf{s}_{ij}\right]_{m\times n}$.

Step 3.2: Determine the deviation of alternatives.

$$d_j\left(\zeta_\alpha,\zeta_\beta\right)=\mathsf{s}_j\left(\zeta_\alpha\right)-\mathsf{s}_j\left(\zeta_\beta\right). \tag{24}$$

Step 3.3: Define the fuzzy preference function. Let ζ_α and ζ_β be two alternatives of the set Υ. The fuzzy preference function $\rho_j\left(\alpha,\beta\right)$ is defined as below.

$$\rho_j\left(\alpha,\beta\right)=\begin{cases}0, & d_j\left(\zeta_\alpha,\zeta_\beta\right)\le 0,\\ \mathsf{s}_j\left(\zeta_\alpha\right)-\mathsf{s}_j\left(\zeta_\beta\right), & d_j\left(\zeta_\alpha,\zeta_\beta\right)>0,\end{cases}\quad j=1,\cdots,n, \tag{25}$$

Table 3 SZN-based rating scale [16]

Linguistic Terms for Evaluations	Linguistic Terms for Reliability	SZNs
Extremely good (EG)	Extremely sure (ES)	(0.85,0.10,0.15)
Very good (VG)	Very sure (VS)	(0.70,0.20,0.30)
Good (G)	Sure (S)	(0.60,0.30,0.40)
Slightly good (SG)	Slightly sure (SS)	(0.55,0.40,0.45)
Fair (F)	Neutral (N)	(0.50,0.40,0.50)
Slightly low (SL)	Slightly unsure (SU)	(0.45,0.40,0.55)
Low (L)	Unsure (U)	(0.40,0.30,0.60)
Very low (VL)	Very unsure (VU)	(0.30,0.20,0.70)
Extremely low (EL)	Extremely unsure (EU)	(0.15,0.10,0.85)

where $\rho_j\ (\alpha, \beta)$ is the outranking intensity denoting that α is superior to β.

Step 3.4: For any pair of alternatives α and β, the overall preference degree (α, β) is determined as follows:

$$(\alpha, \beta) = \sum_{j=1}^{n} {}_j\rho_j\ (\alpha, \beta). \tag{26}$$

Step 3.5: Compute the outgoing flow and incoming flow.

$$^{+}(\alpha) = \frac{1}{m-1} \sum_{\beta=1}^{m} (\alpha, \beta). \tag{27}$$

$$^{-}(\alpha) = \frac{1}{m-1} \sum_{\beta=1}^{m} (\beta, \alpha). \tag{28}$$

Step 3.6: Ranking the alternatives according to the net flow.

$$(\alpha) = {}^{+}(\alpha) - {}^{-}(\alpha)\,. \tag{29}$$

5 Experimental Case Study

5.1 Background

The automotive industry is transitioning toward electrification, intelligence, connection, and sharing, namely the "new four modernizations" strategy. It is essential to promote new industrialization and develop high-quality productive forces. The automobile manufacturing paradigm has been progressively transforming into digitally intelligent manufacturing. The R & D and design of automobiles is shifting from hardware to integrated hardware and software architectures. According to the 2025 Government Work Report in China, implementing "AI +" transformation initiatives stimulates dynamism in the digital economy. The strategic integration of digital technologies with existing manufacturing capabilities and market advantages has accelerated the development of intelligent connected energy vehicles. The Ministry of Industry and Information Technology data reveal that 55.7% of new vehicles sold in H1 2024 are equipped with SAE Level 2 + advanced driver assistance systems. The rate is expected to increase to 65% by 2025. The automotive industry is undergoing a profound revolution driven by AI. In recent years, leading automakers such as Changan, Geely, and BYD have released their intelligentization strategies, positioning AI as a pivotal driver for quality enhancement and industrial upgrading within the sector. Integrating energy vehicles with intelligent and connected technologies is not merely a technological innovation, but a fundamental restructuring of the entire automotive industry ecosystem. It will transform automobiles from simple transportation tools to intelligent mobile spaces. Simultaneously, it has created new business models and a more competitive landscape.

In response to the four new modernizations, leaders in automobile manufacturing enterprises should adjust their leadership style to accommodate the evolving situation within digital transformation. Taking a well-known automobile manufacturing enterprise in China as an example, we analyze leadership competency in driving digital transformation within manufacturing enterprises using the proposed methodology. This automotive manufacturing company is initiating a selection process to identify qualified candidates for a middle management leadership position in its passenger vehicle subsidiary. The company will select one candidate for promotion from among five departments: R&D, Marketing, Manufacturing, Quality Control, and Strategic Operations. Sixteen criteria based on the literature review are presented in Table 4. The details of the candidates are listed in Table 5. We then invite five experts from various fields to evaluate the candidates based on the established criteria. The details of the experts are listed in Table 6. The expert selection process followed rigorous criteria to ensure domain expertise and practical relevance to digital transformation in the manufacturing sector. The panel consisted of five highly qualified professionals with experience in digital leadership. All experts were selected based on a minimum of 10 years of

Table 4 Definition of criteria

Dimension	Criteria	Definition	Reference
Digital strategic vision	Digital leading insight (ζ_1)	Explore opportunities and anticipate potential challenges actively, envisioning the impact of digital trends. Then judge the hour, size up the situation, and take the initiative to make strategic plans to pave the way for enhancing organizational growth.	Memon and Ooi [19]
	Digital vision (ζ_2)	Make forward-looking pre-judgments and align organizational objectives with a keen understanding of the dynamics and uncertainty of the organization in digital contexts.	Shahzad [20]
	Digital strategic deployment (ζ_3)	Formulate a strategic plan according to the enterprise's digital development goals.	Lin [4]
	Digital environment control (ζ_4)	Identify the potential risk and establish strategies in advance for risk management and uncertainty absorption.	Wang et al. [3]
Digital technologies learning	Digital knowledge learning (ζ_5)	Improve the digital literacy of employees by providing digital learning resources, including books, lectures, and courses related to the digital systems closely and the frontier of technology.	Shahzad [20]
	Digital technology choosing (ζ_6)	Select appropriate digital technologies that match the enterprise's business demands.	Zhang and Zheng [21]
	Digital technology implementation (ζ_7)	Design, deploy, and expand new businesses, and enhance business process efficiency through digital technologies.	Wang et al. [3]
	Review and iteration of digital technology (ζ_8)	Review and update based on the application of digital technologies and tools.	Lin [4]
Digital team construction	Digital communication and socialization (ζ_9)	Communication can be carried out through digital media, and effective employee socialization can also be achieved in remote contexts.	Wang et al. [3]
	Digital talent development (ζ_{10})	Emphasize the cultivation of digital knowledge and skills for business team members.	Lin [4]
	Digital empower (ζ_{11})	Foster employees to improve technological proficiency by investing resources, formulating incentive policies, and empowering them to evolve and experiment.	Memon and Ooi [19]
	Form the digital operation team (ζ_{12})	Form new teams with employees who master digital technology and business, or incorporate such members into different business teams.	Xia et al. [32]
Digital ethical awareness	Digital empathy ability (ζ_{13})	The leader can empathize, consider the employee's situation, and help them understand digital transformation.	Li et al. [61]
	Humanized communication and decision-making (ζ_{14})	The leaders will not merely issue orders through digital systems but will also consider problems from the employees' perspective and position, choosing the appropriate time to communicate offline.	Zhang and Zheng [21]
	Digital inclusive counseling (ζ_{15})	Regardless of employees' understanding of digitalization, the leaders will help and encourage them to alleviate their resistance to using digital technology.	Zhang and Zheng [21]
	Emotional balance (ζ_{16})	While using data to evaluate the employee's work, the leaders will also pay attention to their true emotions and inner feelings.	Li et al. [61]

Table 5 Details of the candidates

Candidates	Department	Degree	Experience	Professional work
χ_1	R&D	M.S	5+	With a robust technical expertise in product development and modular design, the candidate demonstrates exceptional skills in addressing complex technical challenges. However, his decision-making process occasionally suffers from excessive attention to granular details, which may hinder operational efficiency.
χ_2	Marketing	M.S	7+	The candidate exhibits strong market sensitivity and demonstrates proficiency in formulating innovative marketing strategies. He possesses effective communication skills, enabling him to influence team members successfully. However, his ability to manage long-term objectives needs further development.
χ_3	Manufacturing	M.S	6+	The candidate demonstrates extensive expertise in production management, with strong operational execution and the ability to optimize processes for efficiency improvement. However, his strategic planning capabilities require further development, particularly in adapting to the dynamic shifts in the industry.
χ_4	Quality Control	M.S	8+	The candidate has rich experience in quality control and strong data analysis skills. He excels at identifying potential risks and possesses a clear, customer-oriented mindset. However, he has relatively little experience in team management, and he needs to strike a balance between quality and cost.
χ_5	Strategic Operations	M.S	7+	The candidate demonstrates robust strategic planning capabilities with great proficiency in resource integration and digital systems management, exhibiting strong decision-making aptitude and effective cross-functional collaboration skills. However, his decision-making processes exhibit extended iteration cycles when implementing new business initiatives.

Table 6 Information of experts

Experts	Degree	Fields	Occupations	Direction	Working experience
ε_1	Ph.D	Academia	Researcher	Production management	17+
ε_2	Ph.D	Academia	Researcher	Digital transformation	10+
ε_3	M.S	Manufacturing enterprises	Manager	Passenger vehicles	16+
ε_4	M.S	Manufacturing enterprises	Senior executive	New energy automobile	23+
ε_5	M.S	Data Service Company	Manager	R&D	11+

relevant experience and direct involvement in digital transformation initiatives, ensuring both the depth of knowledge and practical insights necessary for evaluating leadership competencies in the context of digital transformation. This balanced academic and industry expertise composition provided comprehensive perspectives for assessing the candidates' digital leadership potential.

5.2 Experimental Results of the SZN PROMETHEE II Method

According to the presented methodology, the detailed calculation procedure is as follows.

Stage 1: Generation of the group evaluation matrix.

Step 1.1: The five experts evaluate the candidates based on the criteria outlined in Tables 7, 8, 9, 10, 11. Subsequently, based on the linguistic scale shown in Table 3, we transform the original evaluation matrix into an SZN-based individual evaluation matrix.

Table 7 Evaluation of alternatives for the expert ε_1

	ζ_1	ζ_2	ζ_3	ζ_4	ζ_5	ζ_6	ζ_7	ζ_8
χ_1	(SG,S)	(F,SS)	(F,S)	(G,VS)	(VG,ES)	(G,VS)	(SG,S)	(F,SS)
χ_2	(F,SS)	(G,VS)	(SG,S)	(F,SS)	(F,SS)	(SL,N)	(SG,S)	(F,SS)
χ_3	(SL,N)	(F,S)	(SL,SS)	(SG,S)	(SG,S)	(G,VS)	(VG,ES)	(G,VS)
χ_4	(G,VS)	(SG,S)	(SG,VS)	(VG,VS)	(G,VS)	(SG,S)	(G,VS)	(SG,S)
χ_5	(VG,ES)	(EG,ES)	(EG,VS)	(G,S)	(G,S)	(VG,VS)	(G,S)	(SG,SS)
	ζ_9	ζ_{10}	ζ_{11}	ζ_{12}	ζ_{13}	ζ_{14}	ζ_{15}	ζ_{16}
χ_1	(SL,N)	(F,SS)	(F,SS)	(SL,N)	(F,SS)	(SL,N)	(F,SS)	(F,SS)
χ_2	(VG,ES)	(SG,S)	(G,VS)	(SG,S)	(G,VS)	(VG,ES)	(G,VS)	(G,VS)
χ_3	(F,SS)	(SG,S)	(SG,S)	(F,SS)	(SL,N)	(F,SS)	(SL,N)	(F,SS)
χ_4	(F,SS)	(F,SS)	(F,SS)	(F,SS)	(SG,S)	(SG,S)	(F,SS)	(SG,S)
χ_5	(G,VS)	(VG,VS)	(G,S)	(VG,ES)	(SG,S)	(SG,S)	(SG,S)	(SG,S)

Table 8 Evaluation of alternatives for the expert ε_2

	ζ_1	ζ_2	ζ_3	ζ_4	ζ_5	ζ_6	ζ_7	ζ_8
χ_1	(F,SS)	(SL,N)	(F,SS)	(G,VS)	(EG,ES)	(VG,VS)	(G,VS)	(SG,S)
χ_2	(SG,S)	(G,VS)	(SG,S)	(F,SS)	(F,SS)	(SL,N)	(F,SS)	(SL,N)
χ_3	(SL,N)	(F,SS)	(SL,N)	(SG,S)	(G,VS)	(G,VS)	(VG,ES)	(G,VS)
χ_4	(G,VS)	(SG,S)	(G,VS)	(VG,VS)	(G,VS)	(SG,S)	(G,VS)	(SG,S)
χ_5	(EG,ES)	(EG,ES)	(EG,ES)	(VG,VS)	(G,S)	(VG,VS)	(G,S)	(SG,SS)
	ζ_9	ζ_{10}	ζ_{11}	ζ_{12}	ζ_{13}	ζ_{14}	ζ_{15}	ζ_{16}
χ_1	(SL,N)	(F,SS)	(F,SS)	(SL,N)	(F,SS)	(SL,N)	(F,SS)	(F,SS)
χ_2	(VG,ES)	(G,VS)	(G,VS)	(G,VS)	(G,VS)	(VG,ES)	(G,VS)	(G,VS)
χ_3	(F,SS)	(SG,S)	(SG,S)	(F,SS)	(F,SS)	(F,SS)	(F,SS)	(F,SS)
χ_4	(F,SS)	(F,SS)	(F,SS)	(F,SS)	(SG,S)	(SG,S)	(F,SS)	(SG,S)
χ_5	(G,VS)	(VG,VS)	(G,S)	(VG,ES)	(G,S)	(G,S)	(G,S)	(G,S)

Table 9 Evaluation of alternatives for the expert ε_3

	ζ_1	ζ_2	ζ_3	ζ_4	ζ_5	ζ_6	ζ_7	ζ_8
χ_1	(SG,S)	(F,SS)	(SG,S)	(G,VS)	(EG,ES)	(VG,VS)	(G,VS)	(SG,S)
χ_2	(G,VS)	(SG,S)	(F,SS)	(F,SS)	(F,SS)	(SL,N)	(F,SS)	(SL,N)
χ_3	(F,SS)	(SL,N)	(SL,N)	(SG,S)	(G,VS)	(G,VS)	(VG,VS)	(G,VS)
χ_4	(SG,S)	(F,SS)	(F,SS)	(VG,VS)	(SG,S)	(SG,S)	(SG,S)	(F,SS)
χ_5	(EG,ES)	(VG,VS)	(EG,ES)	(G,VS)	(G,VS)	(VG,VS)	(G,VS)	(SG,S)
	ζ_9	ζ_{10}	ζ_{11}	ζ_{12}	ζ_{13}	ζ_{14}	ζ_{15}	ζ_{16}
χ_1	(SL,N)	(F,SS)	(F,SS)	(SL,N)	(F,SS)	(SL,N)	(F,SS)	(F,SS)
χ_2	(VG,ES)	(G,VS)	(G,VS)	(G,VS)	(G,VS)	(VG,ES)	(G,VS)	(G,VS)
χ_3	(F,SS)	(SG,S)	(SG,S)	(F,SS)	(F,SS)	(F,SS)	(F,SS)	(F,SS)
χ_4	(F,SS)	(F,SS)	(F,SS)	(SL,N)	(SG,S)	(SG,S)	(F,SS)	(SG,S)
χ_5	(G,VS)	(VG,VS)	(G,VS)	(VG,ES)	(G,VS)	(G,VS)	(G,VS)	(G,VS)

Steps 1.2-1.5: First, we determine the positive and negative reference preference data using Eqs. (13) and (14), namely, $M_t^+ = (0.784, 0.122, 0.183)$ and $M_t^- = (0.318, 0.480, 0.629)$. The weights of the experts can then be computed using Eqs. (15)–(18), as listed in Table 12.

Step 1.6: The aggregated group decision matrix is generated using Eq. (19), as presented in Table 13.

Stage 2: Calculation of criteria weights.

Steps 2.1-2.4: First, we calculate the score values of the group decision matrix using Eq. (20). We then normalize these values using Eq. (21), as presented in Table 14. Next, the weights of the competencies are calculated using Eqs. (22) and (23), given in Table 15 and Fig. 3. According to Table 15 and Fig. 3, the competency ζ_{16} "Emotional balance" is

Table 10 Evaluation of alternatives for the expert ε_4

	ζ_1	ζ_2	ζ_3	ζ_4	ζ_5	ζ_6	ζ_7	ζ_8
χ_1	(G,VS)	(SG,S)	(G,VS)	(VG,ES)	(EG,ES)	(VG,ES)	(VG,ES)	(G,VS)
χ_2	(VG,ES)	(G,VS)	(SG,S)	(SG,S)	(SG,S)	(F,SS)	(SG,S)	(F,SS)
χ_3	(SG,S)	(F,SS)	(F,SS)	(G,VS)	(VG,ES)	(VG,ES)	(EG,ES)	(VG,ES)
χ_4	(G,VS)	(SG,S)	(SG,S)	(EG,ES)	(G,VS)	(G,VS)	(G,VS)	(SG,S)
χ_5	(EG,ES)	(EG,ES)	(EG,ES)	(VG,ES)	(VG,ES)	(EG,ES)	(VG,ES)	(G,VS)
	ζ_9	ζ_{10}	ζ_{11}	ζ_{12}	ζ_{13}	ζ_{14}	ζ_{15}	ζ_{16}
χ_1	(SG,S)	(G,VS)	(G,VS)	(SG,S)	(G,VS)	(SG,S)	(G,VS)	(G,VS)
χ_2	(EG,ES)	(VG,ES)	(VG,ES)	(VG,ES)	(VG,ES)	(EG,ES)	(VG,ES)	(VG,ES)
χ_3	(SG,S)	(G,VS)	(G,VS)	(SG,S)	(SG,S)	(SG,S)	(SG,S)	(SG,S)
χ_4	(SG,S)	(SG,S)	(SG,S)	(F,SS)	(G,VS)	(G,VS)	(SG,S)	(G,VS)
χ_5	(VG,ES)	(EG,ES)	(VG,ES)	(EG,ES)	(VG,ES)	(VG,ES)	(VG,ES)	(VG,ES)

Table 11 Evaluation of alternatives for the expert ε_5

	ζ_1	ζ_2	ζ_3	ζ_4	ζ_5	ζ_6	ζ_7	ζ_8
χ_1	(G,VS)	(SG,S)	(G,VS)	(VG,VS)	(EG,ES)	(VG,VS)	(G,VS)	(G,VS)
χ_2	(SG,S)	(F,SS)	(F,SS)	(F,SS)	(F,SS)	(SL,N)	(F,SS)	(SL,N)
χ_3	(F,SS)	(SL,N)	(SL,N)	(SG,S)	(G,VS)	(G,VS)	(VG,VS)	(G,VS)
χ_4	(SG,S)	(F,SS)	(F,SS)	(VG,VS)	(SG,S)	(SG,S)	(SG,S)	(F,SS)
χ_5	(EG,ES)	(VG,VS)	(EG,ES)	(G,VS)	(G,VS)	(VG,VS)	(G,VS)	(SG,S)
	ζ_9	ζ_{10}	ζ_{11}	ζ_{12}	ζ_{13}	ζ_{14}	ζ_{15}	ζ_{16}
χ_1	(F,SS)	(SG,S)	(SG,S)	(F,SS)	(SG,S)	(F,SS)	(SG,S)	(SG,S)
χ_2	(VG,ES)	(G,VS)	(G,VS)	(G,VS)	(G,VS)	(VG,ES)	(G,VS)	(G,VS)
χ_3	(F,SS)	(SG,S)	(SG,S)	(F,SS)	(F,SS)	(F,SS)	(F,SS)	(F,SS)
χ_4	(F,SS)	(F,SS)	(F,SS)	(SL,N)	(SG,S)	(SG,S)	(F,SS)	(SG,S)
χ_5	(G,VS)	(VG,VS)	(G,VS)	(VG,ES)	(G,VS)	(G,VS)	(G,VS)	(G,VS)

Table 12 Weights of experts

	dp_t	dn_t	dr_t	ω_t
ε_1	33.698	15.546	0.684	0.219
ε_2	32.631	17.133	0.656	0.210
ε_3	33.218	16.435	0.669	0.214
ε_4	23.667	27.126	0.466	0.149
ε_5	32.393	17.532	0.649	0.208

the most important, followed by the competencyζ_{13}"Digital empathy ability."

Stage 3: Candidate ranking.

Steps 3.1-3.4: First, we determine the deviation of the candidates using Eq. (24). We then calculate the fuzzy preference values between any two candidates using Eq. (25), as presented in Table 16. Next, we compute the overall preference degree for any pair of candidates using Eq. (26), as presented in Table 17.

Steps 3.5-3.6: We compute the outgoing and incoming flows using Eqs. (27) and (29), as listed in Table 18. Finally, the net flows are calculated, and the candidates are ranked using Eq. (29), as presented in Table 18. From Table 18, the priority of candidates can be exhibited as $\chi_5 > \chi_2 > \chi_1 > \chi_3 > \chi_4$. Namely, the candidate χ_5 has the highest priority, followed by the candidate χ_2, and the candidate χ_4 has the lowest priority.

5.3 Sensitivity Analysis

5.3.1 The Impact of Varying Criteria Weights

In this subsection, we conduct a sensitivity analysis to underline the validity and robustness of the proposed methodology. In reference to Wang et al. [50], we generated 20 different situations for changing criteria weights via random production. The calculation results are shown in Table 19 and Fig. 4.

According to Table 19 and Fig. 4, the random criteria weights can influence the priority of the candidates. According to Fig. 4, the candidate χ_5 has a consistent ranking order among the 20 situations, maintaining the highest priority. Furthermore, the candidate χ_2 maintains the second ranking order in most situations, and the candidate χ_1 occupies

Table 13 Group decision matrix

	χ_1	χ_2	χ_3	χ_4	χ_5
ζ_1	(0.452,0.397,0.497)	(0.496,0.396,0.461)	(0.367,0.472,0.580)	(0.477,0.367,0.471)	(0.761,0.137,0.205)
ζ_2	(0.387,0.465,0.560)	(0.468,0.422,0.481)	(0.357,0.472,0.590)	(0.408,0.458,0.538)	(0.730,0.156,0.234)
ζ_3	(0.447,0.397,0.502)	(0.408,0.504,0.538)	(0.334,0.480,0.614)	(0.437,0.421,0.513)	(0.771,0.131,0.197)
ζ_4	(0.557,0.281,0.396)	(0.384,0.523,0.562)	(0.444,0.417,0.502)	(0.644,0.210,0.313)	(0.551,0.286,0.402)
ζ_5	(0.761,0.137,0.205)	(0.384,0.523,0.562)	(0.527,0.318,0.427)	(0.477,0.367,0.471)	(0.529,0.310,0.425)
ζ_6	(0.585,0.252,0.368)	(0.331,0.560,0.617)	(0.540,0.299,0.414)	(0.444,0.417,0.502)	(0.644,0.210,0.313)
ζ_7	(0.527,0.318,0.427)	(0.396,0.515,0.550)	(0.666,0.197,0.295)	(0.461,0.391,0.486)	(0.529,0.310,0.425)
ζ_8	(0.452,0.397,0.497)	(0.342,0.560,0.605)	(0.540,0.299,0.414)	(0.408,0.458,0.538)	(0.438,0.428,0.508)
ζ_9	(0.356,0.472,0.592)	(0.683,0.236,0.281)	(0.384,0.472,0.562)	(0.384,0.472,0.562)	(0.540,0.299,0.414)
ζ_{10}	(0.416,0.435,0.532)	(0.527,0.367,0.427)	(0.444,0.417,0.502)	(0.384,0.472,0.562)	(0.644,0.210,0.313)
ζ_{11}	(0.416,0.435,0.532)	(0.540,0.355,0.414)	(0.444,0.417,0.502)	(0.384,0.472,0.562)	(0.529,0.310,0.425)
ζ_{12}	(0.356,0.472,0.592)	(0.527,0.367,0.427)	(0.384,0.472,0.562)	(0.361,0.480,0.586)	(0.683,0.188,0.281)
ζ_{13}	(0.416,0.435,0.532)	(0.540,0.355,0.414)	(0.374,0.472,0.573)	(0.444,0.417,0.502)	(0.523,0.323,0.432)
ζ_{14}	(0.356,0.472,0.592)	(0.683,0.236,0.281)	(0.384,0.472,0.562)	(0.444,0.417,0.502)	(0.523,0.323,0.432)
ζ_{15}	(0.416,0.435,0.532)	(0.540,0.355,0.414)	(0.374,0.472,0.573)	(0.384,0.472,0.562)	(0.523,0.323,0.432)
ζ_{16}	(0.416,0.435,0.532)	(0.540,0.355,0.414)	(0.384,0.472,0.562)	(0.444,0.417,0.502)	(0.523,0.323,0.432)

Table 14 Normalized score values of the group decision matrix

	ζ_1	ζ_2	ζ_3	ζ_4	ζ_5	ζ_6	ζ_7	ζ_8	ζ_9	ζ_{10}	ζ_{11}	ζ_{12}	ζ_{13}	ζ_{14}	ζ_{15}	ζ_{16}
χ_1	0.746	0.932	0.756	0.460	0.000	0.389	0.543	0.746	1.000	0.848	0.848	1.000	0.848	1.000	0.848	0.848
χ_2	0.496	0.569	0.774	0.844	0.844	1.000	0.811	0.976	0.000	0.404	0.369	0.404	0.369	0.000	0.369	0.369
χ_3	0.914	0.936	1.000	0.672	0.397	0.353	0.000	0.353	0.872	0.672	0.672	0.872	0.896	0.872	0.896	0.872
χ_4	0.561	0.846	0.731	0.000	0.561	0.700	0.629	0.846	0.928	0.928	0.928	1.000	0.700	0.700	0.928	0.700
χ_5	0.024	0.106	0.000	0.607	0.679	0.336	0.679	1.000	0.644	0.336	0.679	0.235	0.706	0.706	0.706	0.706

Table 15 Calculation of criteria weights

	ζ_1	ζ_2	ζ_3	ζ_4	ζ_5	ζ_6	ζ_7	ζ_8	ζ_9	ζ_{10}	ζ_{11}	ζ_{12}	ζ_{13}	ζ_{14}	ζ_{15}	ζ_{16}
P_j	0.622	0.751	0.660	0.613	0.578	0.749	0.653	1.138	0.648	0.951	1.216	0.768	1.261	0.650	1.221	1.279
$_j$	0.045	0.055	0.048	0.045	0.042	0.054	0.047	0.083	0.047	0.069	0.088	0.056	0.092	0.047	0.089	0.093

the third ranking order in most situations. The candidate χ_3 demonstrates different ranking orders in certain situations, yet exhibits consistent ranking orders in most situations. Specifically, the candidate χ_4 lies in the fifth order in 11 situations, and ranks fourth in other situations. Moreover, the ranking order of the six situations is consistent with the experimental results in this study, namely third, sixth, eighth, thirteenth, seventeenth, and last. Thus, the presented SZN PROMETHEE II methodology is a robust and stable analysis tool for analyzing the leadership competency of manufacturing enterprises undergoing digital transformation.

5.3.2 The Impact of Varying the Number of Experts

The evaluation information experts provide influences the proposed model's computational outcomes. A sensitivity analysis is conducted using varying expert panel sizes to assess the robustness of the results with respect to the number of experts. The number of experts affects the values calculated from Eqs. (12 to 19). Four additional experts—two from academia and two from industry—are invited to evaluate the candidates according to the established criteria. We increase the number of experts sequentially. The calculation results are shown in Table 20 and Fig. 5. According to Table 20 and Fig. 5, the ranking of the candidates remains consistent from $t = 6$ to $t = 9$, which is $\chi_5 > \chi_2 > \chi_1 > \chi_4 > \chi_3$. The Pearson correlation coefficient between the number of

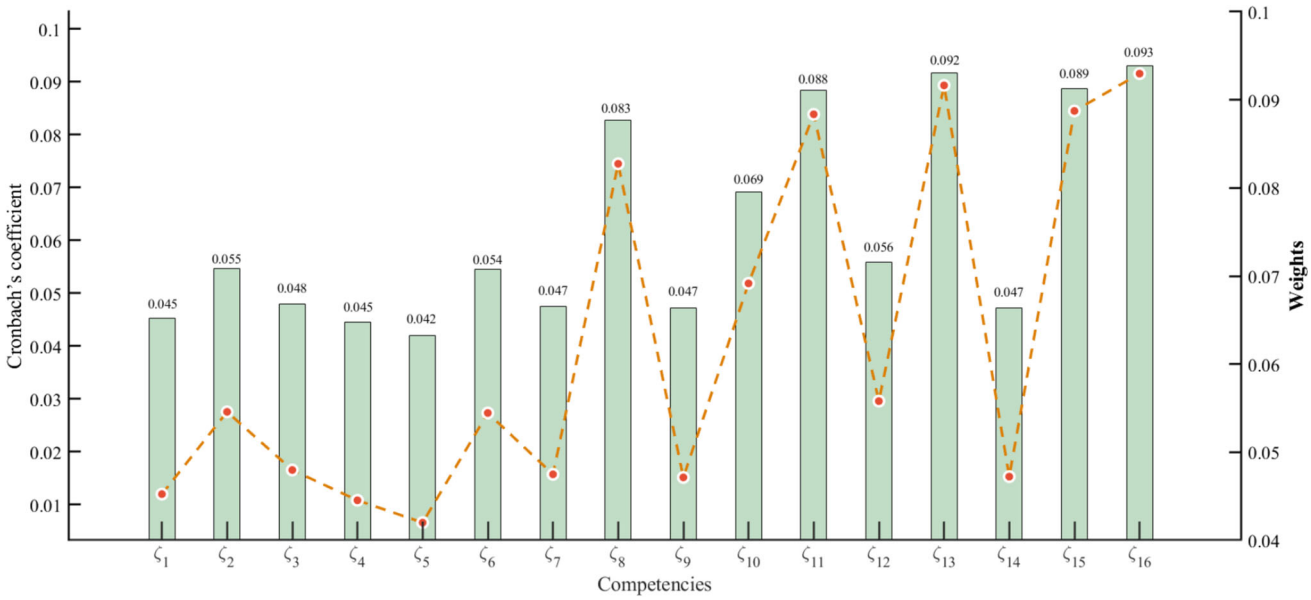

Fig. 3 The weights of competencies

Table 16 Fuzzy preference values between any two candidates

	ζ_1	ζ_2	ζ_3	ζ_4	ζ_5	ζ_6	ζ_7	ζ_8	ζ_9	ζ_{10}	ζ_{11}	ζ_{12}	ζ_{13}	ζ_{14}	ζ_{15}	ζ_{16}
ρ_j (1, 2)	0.000	0.000	0.083	0.259	0.480	0.364	0.203	0.181	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
ρ_j (1, 3)	0.112	0.031	0.142	0.152	0.262	0.056	0.000	0.000	0.000	0.000	0.000	0.000	0.056	0.000	0.056	0.046
ρ_j (1, 4)	0.000	0.000	0.020	0.000	0.323	0.186	0.086	0.066	0.000	0.046	0.046	0.000	0.000	0.000	0.046	0.000
ρ_j (1, 5)	0.000	0.000	0.000	0.007	0.258	0.000	0.000	0.024	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
ρ_j (2, 3)	0.151	0.127	0.059	0.000	0.000	0.000	0.000	0.000	0.362	0.095	0.111	0.176	0.202	0.362	0.202	0.193
ρ_j (2, 4)	0.003	0.071	0.000	0.000	0.000	0.000	0.000	0.000	0.362	0.176	0.193	0.202	0.111	0.280	0.193	0.111
ρ_j (2, 5)	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.151	0.000	0.000	0.000	0.006	0.175	0.006	0.006
ρ_j (3, 4)	0.000	0.000	0.000	0.000	0.062	0.130	0.247	0.182	0.000	0.081	0.081	0.026	0.000	0.000	0.000	0.000
ρ_j (3, 5)	0.000	0.000	0.000	0.000	0.000	0.000	0.157	0.140	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
ρ_j (4, 5)	0.000	0.000	0.000	0.107	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	ζ_1	ζ_2	ζ_3	ζ_4	ζ_5	ζ_6	ζ_7	ζ_8	ζ_9	ζ_{10}	ζ_{11}	ζ_{12}	ζ_{13}	ζ_{14}	ζ_{15}	ζ_{16}
ρ_j (2, 1)	0.039	0.095	0.000	0.000	0.000	0.000	0.000	0.000	0.389	0.130	0.147	0.204	0.147	0.389	0.147	0.147
ρ_j (3, 1)	0.000	0.000	0.000	0.000	0.000	0.000	0.162	0.116	0.027	0.035	0.035	0.027	0.000	0.027	0.000	0.000
ρ_j (4, 1)	0.036	0.024	0.000	0.099	0.000	0.000	0.000	0.000	0.027	0.000	0.000	0.002	0.035	0.109	0.000	0.035
ρ_j (5, 1)	0.360	0.417	0.374	0.000	0.000	0.065	0.004	0.000	0.239	0.286	0.151	0.399	0.141	0.214	0.141	0.141
ρ_j (3, 2)	0.000	0.000	0.000	0.107	0.219	0.308	0.365	0.297	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
ρ_j (4, 2)	0.000	0.000	0.064	0.358	0.157	0.178	0.117	0.115	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
ρ_j (5, 2)	0.321	0.322	0.458	0.251	0.223	0.429	0.207	0.157	0.000	0.156	0.005	0.196	0.000	0.000	0.000	0.000
ρ_j (4, 3)	0.148	0.055	0.123	0.251	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.091	0.081	0.010	0.081
ρ_j (5, 3)	0.472	0.449	0.517	0.144	0.004	0.121	0.000	0.000	0.211	0.251	0.116	0.372	0.197	0.187	0.197	0.187
ρ_j (5, 4)	0.323	0.394	0.394	0.000	0.066	0.251	0.090	0.042	0.211	0.332	0.197	0.398	0.105	0.105	0.187	0.105

Table 17 Overall preference degree for any pair of candidates

	ζ_1	ζ_2	ζ_3	ζ_4	ζ_5	ζ_6	ζ_7	ζ_8	ζ_9	ζ_{10}	ζ_{11}	ζ_{12}	ζ_{13}	ζ_{14}	ζ_{15}	ζ_{16}
$_j(1,2)$	0.000	0.000	0.004	0.012	0.020	0.020	0.010	0.015	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
$_j(1,3)$	0.005	0.002	0.007	0.007	0.011	0.003	0.000	0.000	0.000	0.000	0.000	0.000	0.005	0.000	0.005	0.004
$_j(1,4)$	0.000	0.000	0.001	0.000	0.014	0.010	0.004	0.005	0.000	0.003	0.004	0.000	0.000	0.000	0.004	0.000
$_j(1,5)$	0.000	0.000	0.000	0.000	0.011	0.000	0.000	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
$_j(2,3)$	0.007	0.007	0.003	0.000	0.000	0.000	0.000	0.000	0.017	0.007	0.010	0.010	0.019	0.017	0.018	0.018
$_j(2,4)$	0.000	0.004	0.000	0.000	0.000	0.000	0.000	0.000	0.017	0.012	0.017	0.011	0.010	0.013	0.017	0.010
$_j(2,5)$	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.007	0.000	0.000	0.000	0.001	0.008	0.001	0.001
$_j(3,4)$	0.000	0.000	0.000	0.000	0.003	0.007	0.012	0.015	0.000	0.006	0.007	0.001	0.000	0.000	0.000	0.000
$_j(3,5)$	0.000	0.000	0.000	0.000	0.000	0.000	0.007	0.012	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
$_j(4,5)$	0.000	0.000	0.000	0.005	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	ζ_1	ζ_2	ζ_3	ζ_4	ζ_5	ζ_6	ζ_7	ζ_8	ζ_9	ζ_{10}	ζ_{11}	ζ_{12}	ζ_{13}	ζ_{14}	ζ_{15}	ζ_{16}
$_j(2,1)$	0.002	0.005	0.000	0.000	0.000	0.000	0.000	0.000	0.018	0.009	0.013	0.011	0.013	0.018	0.013	0.014
$_j(3,1)$	0.000	0.000	0.000	0.000	0.000	0.000	0.008	0.010	0.001	0.002	0.003	0.002	0.000	0.001	0.000	0.000
$_j(4,1)$	0.002	0.001	0.000	0.004	0.000	0.000	0.000	0.000	0.001	0.000	0.000	0.000	0.003	0.005	0.000	0.003
$_j(5,1)$	0.016	0.023	0.018	0.000	0.000	0.004	0.000	0.000	0.011	0.020	0.013	0.022	0.013	0.010	0.012	0.013
$_j(3,2)$	0.000	0.000	0.000	0.005	0.009	0.017	0.017	0.025	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
$_j(4,2)$	0.000	0.000	0.003	0.016	0.007	0.010	0.006	0.010	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
$_j(5,2)$	0.015	0.018	0.022	0.011	0.009	0.023	0.010	0.013	0.000	0.011	0.000	0.011	0.000	0.000	0.000	0.000
$_j(4,3)$	0.007	0.003	0.006	0.011	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.008	0.004	0.001	0.008
$_j(5,3)$	0.021	0.024	0.025	0.006	0.000	0.007	0.000	0.000	0.010	0.017	0.010	0.021	0.018	0.009	0.017	0.017
$_j(5,4)$	0.015	0.021	0.019	0.000	0.003	0.014	0.004	0.003	0.010	0.023	0.017	0.022	0.010	0.005	0.017	0.010

Table 18 Results of ranking candidates

	$^+(\alpha)$	$^-(\alpha)$	(α)	Ranking
χ_1	0.047	0.085	−0.038	3
χ_2	0.094	0.087	0.008	2
χ_3	0.042	0.108	−0.066	4
χ_4	0.031	0.100	−0.070	5
χ_5	0.179	0.013	0.165	1

experts used in the case study and the results for $t = 6$ to $t = 9$ is 0.9. The only variation observed is an exchange in priority between candidates χ_3 and χ_4. The candidate with the highest priority is χ_5 for all situations. These findings demonstrate that the proposed model is robust.

5.4 Comparative Analysis

In this subsection, we employ a comparative analysis with different SZN-based MCDM methods to examine the applicability of the results obtained in this study. It includes two phases. First, we select the MCDM methods utilized in competency evaluation research, encompassing the TOPSIS, COPRAS, WASPAS, and EDAS methods. The PROMETHEE I method is also chosen to handle the same case of the study. Next, we employ the existing SZN-based MCDM methods in the literature, including the prospect theory, TODIM, ARAS, and the MARCOS methods.

5.4.1 Comparison of Existing Competency Analysis Methods

The results of the comparative analysis with existing competency analysis methods are presented in Tables 21 and 22, Fig. 6. According to Table 21 and Fig. 6, the priorities of the candidates calculated by SZN TOPSIS and SZN COPRAS are similar, which are $\chi_4 > \chi_3 > \chi_1 > \chi_2 > \chi_5$ and $\chi_3 > \chi_4 > \chi_1 > \chi_2 > \chi_5$. Namely, only the ranking of candidates χ_3 and $\chi_4$is exchanged. It is significantly different from the proposed methodology. The TOPSIS method is based on the principle of selecting the alternative that simultaneously minimizes the distance to the positive ideal solution and maximizes the distance from the negative ideal solution. However, in practice, the ideal solution may not correspond to any actual alternative, which can result in biased or misleading rankings [62]. The COPRAS method ranks alternatives by explicitly distinguishing between competing criteria, such as those categorized as beneficial or costly [63]. However, in the constructed competency index, all the competencies fall under the benefit criteria. Thus, the COPRAS method is not appropriate for dealing with this problem. The priority of can-

Table 19 Results of the sensitivity analysis on varying criteria weights

	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
χ_1	−0.030	−0.005	−0.029	−0.075	−0.005	−0.028	−0.032	−0.013	−0.041	−0.014
χ_2	−0.007	−0.069	−0.013	0.074	−0.029	−0.009	−0.001	−0.010	−0.013	−0.040
χ_3	−0.074	−0.032	−0.069	−0.097	−0.031	−0.056	−0.068	−0.034	−0.055	−0.046
χ_4	−0.058	−0.058	−0.069	−0.071	−0.046	−0.070	−0.062	−0.053	−0.052	−0.048
χ_5	0.169	0.164	0.180	0.170	0.111	0.164	0.163	0.110	0.161	0.148
	11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
χ_1	0.013	−0.038	−0.034	−0.024	−0.019	−0.007	−0.026	0.013	−0.001	−0.020
χ_2	−0.064	−0.013	−0.003	0.022	−0.001	−0.034	−0.013	−0.062	−0.048	−0.005
χ_3	−0.051	−0.038	−0.051	−0.079	−0.069	−0.062	−0.037	−0.040	−0.031	−0.043
χ_4	−0.056	−0.073	−0.071	−0.050	−0.058	−0.079	−0.072	−0.058	−0.086	−0.053
χ_5	0.159	0.162	0.158	0.131	0.147	0.182	0.148	0.147	0.166	0.121

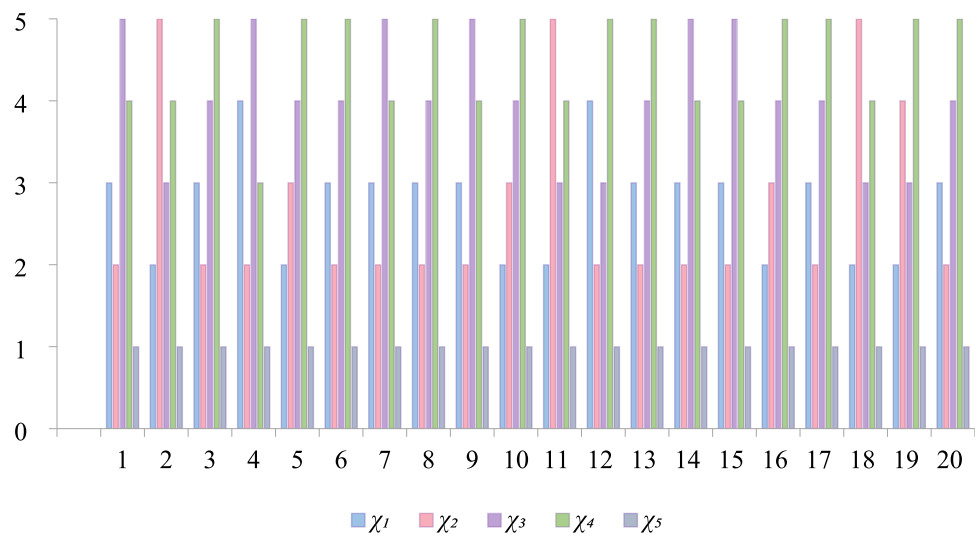

Fig. 4 Results of the sensitivity analysis on varying criteria weights

Table 20 Results of sensitivity analysis on the number of experts

	$t = 5$		$t = 6$		$t = 7$		$t = 8$		$t = 9$	
	(α)	Ranking	(α)	Ranking	(α)	Ranking	(α)	Ranking	(α)	Ranking
χ_1	−0.038	3	−0.344	3	−0.472	3	−0.462	3	−0.503	3
χ_2	0.008	2	0.121	2	0.413	2	0.469	2	0.288	2
χ_3	−0.066	4	−0.654	5	−0.679	5	−0.747	5	−0.794	5
χ_4	−0.070	5	−0.572	4	−0.549	4	−0.564	4	−0.539	4
χ_5	0.165	1	1.448	1	1.287	1	1.304	1	1.548	1

didates acquired by the SZN WASPAS method is $\chi_5 > \chi_2 > \chi_1 > \chi_4 > \chi_3$, which is similar to the proposed methodology. The value of the Pearson correlation coefficient between the SZN WASPAS method and the proposed methodology is 0.9. Furthermore, the priority of candidates employed by the SZN EDAS method is consistent with the proposed method, which is $\chi_5 > \chi_2 > \chi_1 > \chi_3 > \chi_4$. The total ranking order is not obtained using the SZN PROMETHEE II method due to its partial ranking characteristic. It can only find that the candidate χ_5 has the highest priority among the candidates, which is consistent with the presented model.

According to Table 22, the proposed SZN PROMETHEE II can model the experts' preferences using the preference functions compared with other MCDM methods, especially within fuzzy settings. Further, the proposed model offers a complete ranking from best to worst, rather than the partial

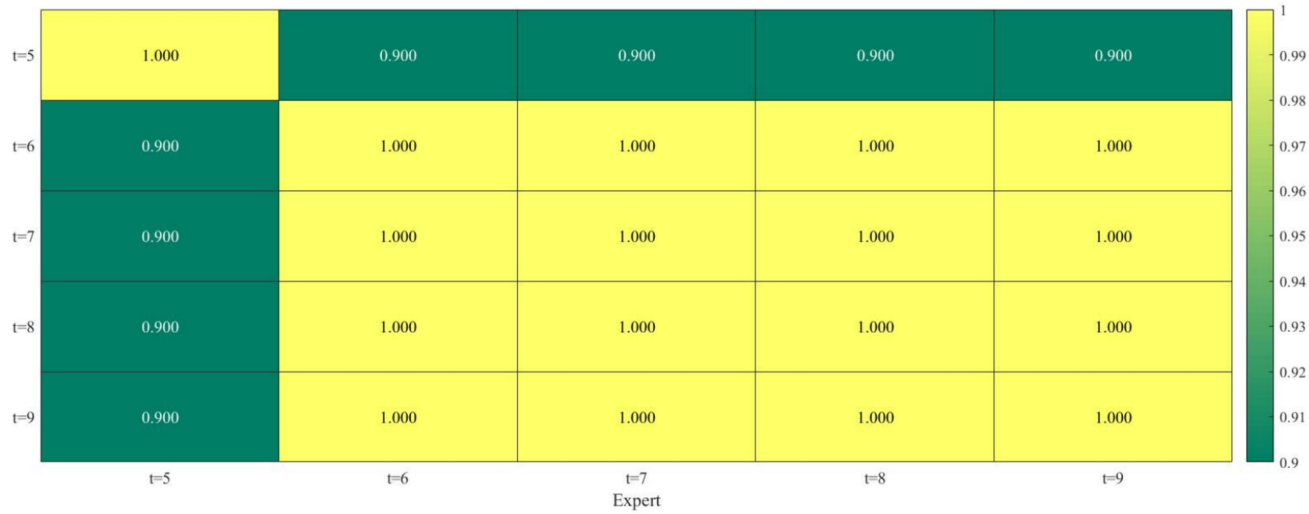

Fig. 5 Values of the Pearson correlation coefficient for different numbers of experts

pre-order for the SZN PROMETHEE II method. The methodology provides high outranking clarity, outperforming SZN TOPSIS, SZN COPRAS, SZN WASPAS, and SZN EDAS methods. Therefore, the proposed model demonstrates strong robustness, establishing it as a reliable and effective tool for evaluating the leadership competencies of manufacturing enterprises in the context of digital transformation.

5.4.2 Comparison of Existing SZN-Based Methods

The results of the comparative analysis with existing SZN-based methods are presented in Table 23 and Fig. 7. According to Table 23 and Fig. 7, the priorities of the candidates computed by the SZN prospect theory and SZN TODIM are significantly different compared to the proposed method, which are $\chi_3 > \chi_4 > \chi_1 > \chi_2 > \chi_5$ and $\chi_4 > \chi_5 > \chi_1 > \chi_2 > \chi_3$, respectively. These two methods are non-compensatory, while others are compensatory or partially compensatory methods. The priority of candidates conducted by the SZN MARCOS is consistent with our proposed method, and the value of the Pearson correlation coefficient between them is 1. Hence, our method shows robustness and provides a reliable analysis of leadership competencies in digitally transforming manufacturing enterprises.

5.5 Managerial Implications

This study provides valuable insight for the organization on the appraisal of leadership competency within digital transformation. Several managerial implications are given in this subsection:

- Leaders should maintain an open and inclusive attitude, encouraging organizational members to explore and innovate. However, a balance between managerial automation and algorithmic processes should be maintained, ensuring employee emotional fulfillment and recognition of value. It is necessary to avoid controlling employees rigorously in a completely rational manner.
- Leaders should have a forward-looking digital strategic awareness, attaching importance to sustainable digitalization. Furthermore, they should proactively embrace digital transformation and innovation, striving to promote digital transformation by optimizing the rational allocation of human, material, and financial resources.
- Leaders should regularly monitor and evaluate the application of digital technologies in enterprises, accelerating the elimination of outdated and inefficient equipment. They should continuously iterate and improve the application of digital technologies in response to the evolving business needs of enterprises.

6 Conclusion

Analyzing leadership competency in driving digital transformation is crucial for manufacturing enterprises to adapt to the evolving digital landscape. However, little attention has been given to this topic from the perspective of decision-making methods. Hence, this study aims to introduce an integrated MCDM method, the SZN PROMETHEE II methodology, to analyze leadership competency in driving digital transformation for manufacturing enterprises. First, the SZN RCC method, utilizing the OWA operator, aggregates information from experts. Compared to existing fuzzy sets, SZN encompasses more uncertain information and considers the experts'

Table 21 Results of the comparative analysis with existing competency evaluation methods

	SZN TOPSIS		SZN COPRAS		SZN WASPAS		SZN EDAS		SZN PROMETHEE I			Proposed methodology
	Value	Ranking	Value	Ranking	Value	Ranking	Value	Ranking	$^{+}(\alpha)$	$^{-}(\alpha)$	Ranking	Ranking
χ_1	0.411	3	0.933	3	7.742	3	0.141	3	0.047	0.085	–	3
χ_2	0.500	4	0.873	4	7.761	2	0.281	2	0.094	0.087	–	2
χ_3	0.343	2	1.000	1	7.691	5	0.064	4	0.042	0.108	–	4
χ_4	0.320	1	0.997	2	7.703	4	0.040	5	0.031	0.100	–	5
χ_5	0.750	5	0.653	5	7.998	1	0.924	1	0.179	0.013	–	1

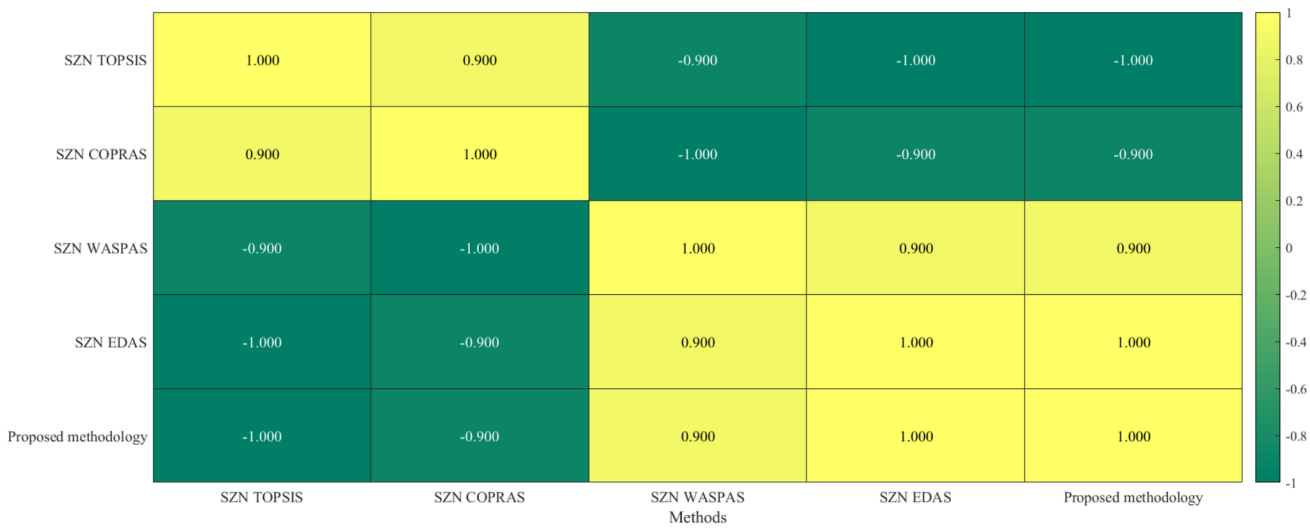

Fig. 6 Values of the Pearson correlation coefficient with existing competency evaluation methods

Table 22 Comparison of MCDM Methods

Evaluation criterion	SZN TOPSIS	SZN COPRAS	SZN WASPAS	SZN EDAS	SZN PROMETHEE I	Proposed methodology
Preference Modeling	Medium	Medium	Medium	Medium	High	High
Outranking Clarity	Medium	Medium	Medium	Medium	High	High
Completeness of Ranking	High	High	High	High	Low	High

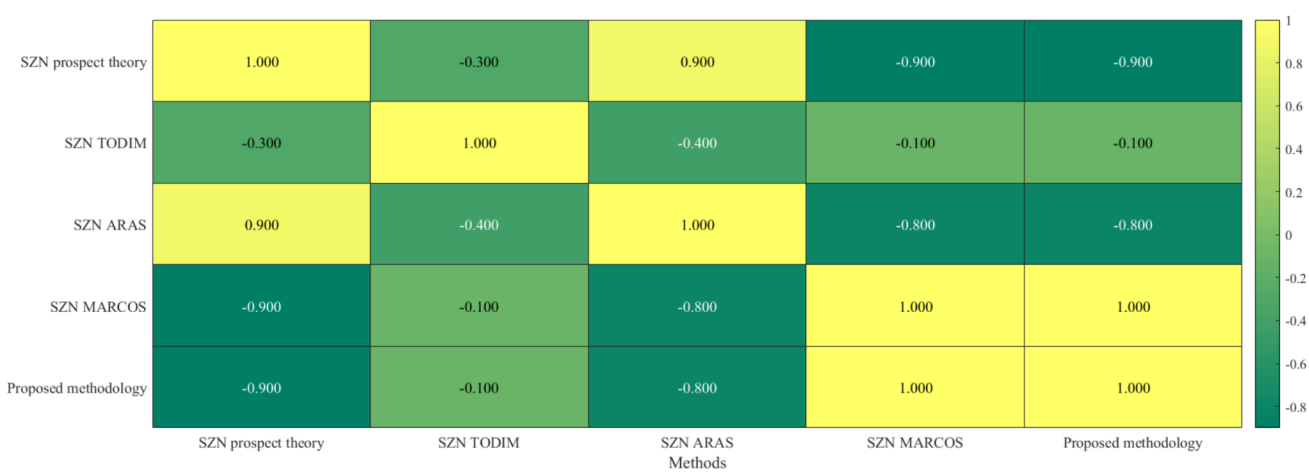

Fig. 7 Values of the Pearson correlation coefficient with existing SZN-based methods

Table 23 Results of the comparative analysis with existing SZN-based methods

	SZN prospect theory		SZN TODIM		SZN ARAS		SZN MARCOS		Proposed methodology
	Value	Ranking	Value	Ranking	Value	Ranking	Value	Ranking	Ranking
χ_1	0.429	3	−0.757	3	1.128	4	0.973	3	3
χ_2	0.427	4	−0.884	4	1.131	3	1.021	2	2
χ_3	0.467	1	−1.121	5	1.164	1	0.945	4	4
χ_4	0.459	2	0.024	1	1.143	2	0.945	5	5
χ_5	0.275	5	−0.106	2	1.005	5	1.157	1	1

reliability. Second, SZN LOPCOW is applied to compute the criteria weights, addressing the structural inconsistencies of the criteria. Third, the SZN PROMETHEE II for ranking candidates measures the deviation of any pair of candidates using a simplified counting procedure. It also derives the complete ranking of candidates rather than a partial ranking using the PROMETHEE I method. Subsequently, a comprehensive index system based on sixteen criteria within four aspects is constructed. Then, an automobile manufacturing enterprise in China is used to demonstrate the methodology's feasibility. In the given experimental case study, "Emotional balance (ζ_{16})" with a value of 0.093 is ranked as the most important leadership competency for manufacturing enterprises during digital transformation. Therefore, leaders should emphasize the emotional balance in digital transformation, ensuring that managerial automation and algorithmic processes coexist with employee emotional fulfillment and value recognition. Further, they should avoid rigid rational control, as effective leadership hinges on harmonizing efficiency with human-centric engagement. Then, the competency ζ_{13}"Digital empathy ability" is ranked as the second crucial leadership competency in driving digital transformation within manufacturing enterprises. Digital empathy is essential for leaders steering digital transformation, ensuring human-centric progress while optimizing efficiency and well-being. Under the competencies analysis, the candidate χ_5 has the highest priority, and the candidate χ_4 has the lowest priority. To test the robustness of the proposed methodology, a sensitivity analysis is performed using a set of 20 criterion weight scenarios and by varying the number of experts. Furthermore, a comparative analysis with two phases is conducted to investigate the availability of this study in comparison with prior competency analysis methods and existing SZN-based methods.

However, this study has several limitations. First, the expert sampling methodology lacks sufficient breadth to ensure representative evaluation outcomes across the wider expert population. Second, the potential interrelation of the expert is not considered, which is a common practice situation. Third, the current methodology does not consider two-dimensional information during evaluation, which does not reflect the dynamic characteristics. Therefore, future studies should focus on the following points. First, future studies could expand both the size and diversity of expert panels to improve the representativeness and generalizability of the analysis results. Second, other operators, such as the power geometric operator, can be employed to aggregate information. Third, complex fuzzy sets, such as complex spherical fuzzy sets, can be applied to describe information.

Author contributions Quanguo Bi contributed to methodology, formal analysis, writing—original draft, and funding acquisition. Yushuo Cao contributed to validation, funding acquisition, writing—review & editing, and supervision. Jin Xu contributed to formal analysis and writing—original draft. Hao Wang contributed to validation and writing—review & editing. Weizhong Wang contributed to validation, writing—review & editing, and supervision.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with animals performed by any of the authors.

Informed consent Informed consent was obtained from all individual participants included in the study.

Data transparency All the data have been provided in the manuscript.

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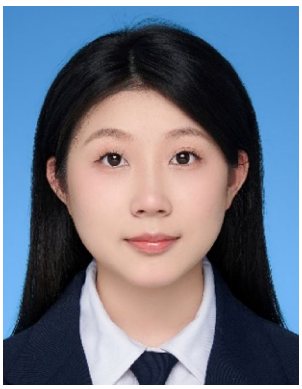

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